



Nunavik Nickel Project
Annual report GHG objectives and emissions - 2023



INTRODUCTION

Canadian Royalties Inc (CRI) mines nickel and copper ores at its Nunavik Nickel Project (NNiP) mining complex. This mining project is located approximately 80 km west of Kangiqsujaq and about 140 km southeast of Salluit in Northern Quebec. NNiP began in 2008 with the discovery of the Mesamax, Expo, Méquillon and Ivakkak deposits. The Allammaq and Puimajuq deposits were subsequently added to the Global Certificate of Authorization No. 3215-14-007 issued by the Quebec government.

An Environmental and Social Impact Assessment (ESIA) was carried out in 2007 to characterize the receiving environment and assess the potential impact of NNiP on it. An Environmental Monitoring Program (EMP) and a set of mitigation measures have been developed and implemented for several years.

As a member of the Quebec Mining Association, CRI adheres to the *Mining* Association of Canada's *Towards Sustainable Mining* (TSM) initiative. CRI implements the Climate Change Protocol by integrating it into its activities and NNiP's Environmental Management Program. CRI has also adopted a climate strategy, presented in the appendix to this report, to guide and structure its actions.

The purpose of this document is to report on CRI's performance in terms of energy management and greenhouse gas (GHG) emissions. This performance should be read in the context of NNiP's isolated northern environment. There is no electrical grid available; all the energy required for operations is generated, for the time being, by burning diesel. However, the deployment of two wind turbines is planned for 2026, in line with the company's climate strategy.

A handwritten signature in blue ink, reading "Stéphane Twigg".

Stéphane Twigg, Environmental Director

RESULTS

1. Summary of GHG emissions

NNiP's GHG emissions are reported in accordance with the *Règlement sur la déclaration obligatoire des émissions de certains contaminants atmosphériques* (RDOCECA), and are subject to external verification by an accredited firm. CRI also participates in the MELCCFP cap-and-trade system (SPEDE). Table 1 shows the emissions reported under the RDOCECA and calculated according to the methodologies prescribed by the latter. These emissions also represent total emissions for scopes 1 and 2. Emissions intensity based on ore processed at the concentrator has been retained as an indicator for 2023.

Table 1: Emissions reported under the RDOCECA in 2023

Types of emissions	Emissions (t CO ₂ eq)	% of total emissions
Electricity generation	79 822	61
Combustion using stationary equipment	17 132	13
Combustion using mobile equipment	32 755	25
Non-specific protocol - Explosives and burning of residual materials at LEMN	704	0,53
Non-specific protocol - Refrigerant gas	61	0,047
TOTAL	130 475	100
Production intensity for ore processing	0.0794 t CO ₂ eq/t ore	

Table 2 shows the carbon footprint of NNiP's copper and nickel production according to the methodology of the *Nickel Institute's* guide [202204-lme-passport-guidance-nickel-institute.pdf](https://nickelinstitute.org/202204-lme-passport-guidance-nickel-institute.pdf) (nickelinstitute.org). This footprint is based on data from 2021, the year in which a scope 3 emissions inventory and life-cycle assessment were conducted. It is expected that the carbon footprint will have remained similar in 2023 compared with 2021 due to the similarity of activities between these two years.

Table 2: Carbon footprint according to *Nickel Institute* methodology

Types of emissions	Emissions (t CO ₂ eq)
Scope 1 - direct on-site emissions for diesel ¹	107 000
Scope 1 - direct on-site emissions for other sources (propane, refrigerant gas, acetylene)	23
Scope 3 - raw materials (including explosives)	40 900
Scope 3 - diesel production	16 104
TOTAL	164 027

¹ Emissions from the Expo camp are excluded according to the Guide's methodology.

2. Energy management targets and objectives

Targets and objectives have been set for energy management, with a view to avoiding energy waste and limiting GHG emissions.

2.1 Energy consumption target for the ore milling sector

The established energy consumption target is 22.5 kWh/t of ore milled. The average consumption for 2023 was 20.61 kWh/t of ore. The target was met for 2023.

2.2 Progress towards 2023 targets

Energy management targets for 2023 included the implementation of the following efficiency measures:

- Replacement of heat exchangers at Centrale Energy to enable better transfer of residual heat from the generators to the glycol network.
- Installation of the Synapse system on part of the Expo camp: this electrical load management system prevents peaks in demand and helps keep generators in their optimum operating range.
- Installation of a continuous remote monitoring system on the generators at the Méquillon site: this system will enable remote monitoring of the generators, enabling Energy Department staff to detect diesel consumption anomalies more quickly and keep the generators within their optimum operating range.

2.3 2024 targets

The main energy management objectives for 2024 are :

- Establish energy efficiency and/or GHG emission targets for key NNiP sectors
- Connecting the camp's glycol network to the concentrator's network, representing a reduction of 937 t CO₂ eq/year
- Continue feasibility study for energy storage unit at Expo site
- Enhanced training for Centrale Energy operators to increase awareness and understanding of the factors influencing energy performance, with a view to optimizing performance for significant energy uses or emission sources.

APPENDIX: CRI'S CLIMATE STRATEGY

Climate strategy

Canadian Royalties Inc. is a mining company committed to protecting the environment and respecting communities. We act to minimize hazards, risks and environmental impacts, and seek to continuously improve our performance. Canadian Royalties Inc. recognizes the importance of combating and adapting to climate change. In keeping with the commitments of its Social and Environmental Responsibility Policy, Canadian Royalties inc. is committed to:

- Manage the energy consumption of its activities and limit its greenhouse gas (GHG) emissions
- Limit the impact of our activities on climate change and adapt to it

Canadian Royalties Inc.'s strategy for meeting these commitments is :

Develop governance, accountability and responsibilities for the following areas:		
Managing climate risks and opportunities	Improving resilience	Reducing carbon footprint
▪ Assess short-, medium- and long-term risks climate change and opportunities at company and facility level ▪ Establish a process for identifying significant risks and quantifying their material and financial impact ▪ Integrate the impacts of climate risks and opportunities into corporate strategy, business processes and financial planning for the various spheres of the Nunavik Nickel Project (mining, value chain, access to capital) ▪ Develop and implement climate change adaptation pathways	▪ Developing tools for adapting to climate change ▪ Consider at least two climate-related scenarios in risk assessment, one based on high carbon emissions and the other on ambitious carbon emission reductions ▪ Collaborate with communities of interest on climate change management ▪ Planning the transition to a low-carbon economy	▪ Formalize an energy and GHG management system, including a monitoring plan, reduction targets for energy consumption and GHGs (scope 1 and 2), procedures and a continuous improvement process for energy efficiency ▪ Work with value chain partners to set Scope 3 emission reduction targets ▪ Assign an implicit cost to carbon in project evaluations and annual budgeting, and integrate it into procurement criteria. ▪ Replace fossil fuels with low-emission energy sources wherever possible ▪ Develop carbon footprint assessment tools based on the principles of life cycle assessment



James Xiang
President/Chief Operating Officer
Canadian Royalties Inc. January
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